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Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022

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A B S T R A C T

In intensively managed poultry production systems the term 'smothering' refers to deaths from suffocation that occur as a consequence of piling behaviour where birds crowd together into densely packed groups. Smothering is a non-negligible source of loss in free-range layer hens, having both negative welfare and economic effects. Smothering events are rarely observed and are usually detected by the discovery of groups of dead hens. The aim of this study was to identify risk factors for smothering deaths in three commercial free-range layer poultry farms in Australia. This was a prospective cohort study of poultry flocks managed by three commercial free-range layer farms in eastern Australia. Flocks were enrolled into the study from 1 January 2019–29 March 2021 and were followed until the end of lay or until the end of the study on 31 March 2022, whichever occurred first. Throughout the follow-up period of the study, daily production and weather data, details of flock management and details of the place and time of smothering events were recorded. Time to event (survival) analyses were used to quantify the association between hypothesised risk factors and the number of days in lay at the time of smothering. Shed and bird level characteristics associated with time to event were quantified using a stratified Cox proportional hazards model which included a frailty term to account for birds clustered within sheds within farm. Across the three farms, for every 100 birds placed into a shed, there were 12 deaths over the duration of the production period. Of the 12 deaths per 100 birds, 2 were due to smothering. Our Cox proportional hazards regression analyses showed that the daily hazard of smothering was increased for birds housed in aviary sheds compared with flat-deck sheds (HR 4.0, 95 % CI 1.7–9.7). The daily hazard of smothering mortality was increased on warm, humid and rainy days, and in birds with low fear of humans and high fear of novel objects. Rainy days on which outdoor daily average humidity was greater than or equal to 70 % were associated with a 3.7 (95 % CI 3.5–3.9) fold increase in the daily hazard of indoor smothering deaths, compared with days when outdoor daily average humidity was less than 70 % and no rain. This study provides useful insight into the determinants of smothering in Australian free-range layer hens, in particular risk factors that do not change over time (e.g., shed type) and those that change daily (e.g., weather conditions). This information allows flock management strategies to be adapted accordingly.

1. Introduction

In intensively managed poultry production systems the term 'smothering' refers to deaths from suffocation that occur as a consequence of piling behaviour where birds crowd together into densely packed groups (Bright and Johnson, 2011a). Smothering is an important economic as well as an animal welfare issue (Rayner et al., 2016; Bright and Johnson, 2011a).

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The small number of published studies documenting the frequency of and risk factors for smothering in free-range layer hens are based on relatively small numbers of flocks with details of the incidence of smothering deaths often obtained by flock manager interviews. A cross-sectional study of 10 commercial layer flocks in the United Kingdom by Bright and Johnson (2011a) was, to the best of our knowledge, the first to quantify the incidence of smothering in free-range layer flocks and to raise the issue of smothering as an animal welfare concern. In their study, Bright and Johnson (2011a) defined three types of smothering: (1) panic; (2) nest box; and (3) recurring or creeping. The definition of panic smothering employed in that study closely resembled the concept of 'panic' as discussed in the review on panic and hysteria in domestic fowl by Mills and Faure (1990). Panic smothers tend to occur without warning and are believed to stem from disturbances that generate a diffusion of panic or terror across the entire flock or a subset of the flock (Gray et al., 2020). Armstrong et al. (2023) and Richards et al. (2012) also reported episodes of intense fear and anxiety as 'hysteria' or 'panic' which can then result in smothering deaths that are distinct from the deaths that occur with creeping piling events. Nest box smothering occurs when a bird visits a nesting box to lay and other birds choose the same nesting box. This could lead to a scenario where multiple birds try to use the nest box simultaneously, potentially resulting in overcrowding and smothering. Creeping or recurring smothering, as implied by its name, refers to a situation where smothering incidents reoccur at various points throughout the laying period (Bright and Johnson, 2011a). Creeping smothering is unpredictable, and there is very little known about why birds engage in this behaviour. Mortalities resulting from creeping smothering can occur both inside and outside the laying house (Bright and Johnson, 2011a).

Rayner et al. (2016) documented correlations between layer housing, disease, management practices and smothering. This questionnaire-based study reported that out of 205 farms, 134 farms experienced smothering events. Rayner et al. (2016) found that breed, nest box manufacturer and increased use of the outdoor range on sunny days were associated with an increased risk of smothering deaths throughout the laying period. They suggested that the variation in nest box design have implications not only for nesting behaviour but also for the overall spatial layout of the shed and such design variations can lead to hens congregating in specific areas, thereby increasing the risk of smothering incidents. The provision of additional oyster grit or grain on the litter has been shown to reduce panic and recurrent smothering among laying hens (Rayner et al., 2016). Reyner et al. suggested that this strategy, aimed at providing rewarding foraging opportunities, serves as an enhancement to regular management practices which may keep birds occupied and reduce unwanted behaviours such as smothering (Rayner et al., 2016) and feather pecking (Lambton et al., 2010).

Smothering deaths are always preceded by piling but piling events do not always result in smothering (Campbell et al., 2016). The threshold at which a piling event results in the death of one or more birds due to smothering is currently unknown. The study by Winter et al. (2021) showed that 78 % of piling events in Swiss layer flocks were preceded by individual hen activities such as standing, sitting and pecking at objects and piling frequency decreased as flocks age. Additionally, mass movement of hens, sunlight spots, stockpersons entering the shed, light spots from lamps, and aggression between hens accounted for 7.6 %, 2.6 %, 1.7 %, 0.9 %, and 0.3 % of piling events, respectively.

An insight from Winter et al. (2022) highlights the multifaceted nature of triggers for piling and smothering in layer flocks, as perceived by British flock managers. Through interviews with twelve flock managers, various events were identified as potential smothering triggers, including the transition from rearing to laying environments, flight responses, disruptions in routine, gregarious nesting behaviour both in nest boxes and on the floor, and other communal activities such as dustbathing. Furthermore, this study showed that the causes of piling and smothering evolve over the flock cycle and vary according to the time of day, underscoring the dynamic nature of these phenomena in commercial poultry farming.

The exact mechanisms that lead to piling events in poultry are poorly understood which means that the observed occurrence of piling (and therefore possibly smothering) events may appear to occur at random. However, there may be environmental factors that are not immediately apparent by humans that could influence the propensity to pile. For example, birds can detect differences in air quality, such as high concentrations of ammonia (Pokharel et al., 2017) and can detect auditory stimuli outside of the range detectable by humans (Hill et al., 2014). Research shows that the effect of light on laying hens is influenced by various factors including light colour, intensity, photoperiod, and light source (Manser, 2023). High-illumination lighting has been found to be associated with negative effects on hen behaviour and welfare, including increased irritability, fearfulness, and activity levels, leading to elevated fighting behaviour and pecking issues (O'Connor et al., 2011; Xie et al., 2008a; Xie et al., 2008b). Moreover, it can result in a decline in eggshell quality, an increase in the number of deformed eggs, and higher mortality rates (Wei et al., 2020; Liu et al., 2018).

Based on the relatively small number of observational studies that have investigated piling and smothering in free-range hens, it appears that piling and smothering, similar to many production diseases of animals, is a multi-factorial problem arising from characteristics operating at the level of the bird (Rayner et al., 2016; de Haas et al., 2013), flock (Winter et al., 2022) and shed (Chowdhury et al., 2024) as well as aspects of bird behaviour, for example, fear and panic responses (Winter et al., 2022; Richards et al., 2012) in addition to the interactions that occur between birds and those that manage them. While it is difficult to make definitive inferences from the studies cited above due to differences in case definitions and study design, they provide a useful list of risk factors for smothering that might be tested using empirical data. With this background, this study aimed to identify bird, flock and shed level characteristics associated with an increase in daily smothering risk in Australian free-range layer hens. The findings from this study provides an evidence-based approach for the design of interventions to reduce the incidence of smothering in free-range layer flocks in Australia.

2. Materials and methods

Ethics approval for this study was provided by the University of XXX Ethics Committee (University of XXX Animal Ethics approval 1814708.1). (Authors' identifiers have been retracted in accordance with journal guidelines).

2.1. Study population

This was a prospective cohort study carried out over three years (January 2019 to March 2022) of 84 flocks of adult laying hens housed across 43 sheds operated by three organisations located in northeastern Victoria ($n = 1$) and southeastern Queensland ($n = 2$). The selection of organisations and farms within each organisation was purposive, primarily guided by the interest of farm managers and their demonstrated capacity to consistently record the necessary variables for each flock on a daily basis throughout the designated follow-up period. In the remainder of this paper we use the terms 'organisation' and 'farm' interchangeably. Flocks were enrolled into the study between 1 January 2019 and 29 March 2021 and were monitored until the end of their production cycle or until 31 March 2022, whichever occurred first. Farm sizes ranged from 3.4 to 43.5 ha, each containing between 17 and 35 sheds used for housing birds. Shed sizes varied from approximately 1265 m^2 to 3100 m^2, with each shed accommodating, on average, between 16,000 and 45,000 birds. Each of the sheds on each of the three study farms were tunnel ventilated. The climate in southeastern Queensland is subtropical, characterised by warm summers and mild winters. The climate in Victoria is temperate maritime, characterised by mild temperatures throughout the year and more rainfall in winter than summer.

Details of the rearing and production systems and management for these flocks are provided by Chowdhury et al. (2024). Briefly, the three organisations used two types of production sheds: flat deck ($n = 20$) and aviary ($n = 23$). Flat deck sheds are single-tier structures with either partially slatted or fully slatted flooring. Some flat deck sheds had an outdoor littered area accessible during the day, known as a 'winter garden' ($n = 9$). Aviary sheds are comprised of multiple levels each with nest boxes, perches, feed and water. Two types of aviary sheds were used by the organisations that took part in this study: A-frame ($n = 14$) and terrace ($n = 9$). A-frame aviaries are central multi-tiered structures, while terrace aviaries consist of multiple structures with channels between them. All study flocks were free-range and had access to the outdoor range area during the daytime from the time the pop holes were opened in the morning until they were closed in the afternoon, except during inclement weather (as detailed in Chowdhury et al., 2024).

We defined a flock as a group of hens placed into the same production shed on the same date (or within one week) and subjected to the same management practices for the duration of their period in lay. A shed was defined as the physical facility in which flocks were housed for the duration of their period in lay. As such, more than one flock may be housed in a shed (over time) however only one shed was allocated to each flock.

Flock managers provided routinely recorded production details for each flock entering the production cycle. Data for each flock were reported for each day of the follow-up period which we defined as the interval from the date of flock enrolment until the end of lay or until the end of the study, whichever occurred first. The study follow-up period for individual flocks ranged from 326 to 503 days.

Flock-level data included the date on which hens were placed into the production shed (at approximately 17 weeks of age), a numeric code to uniquely identify the shed of placement, shed type (aviary, flat deck), the number of staff engaged in flock management on a daily basis and treatment history, feeder space per bird, the number of drinkers per bird, pop hole length per bird and breed (Hy-line Brown, ISA Brown or mixed). The time pop holes were opened and closed, and the time of scheduled stockperson flock checks were recorded daily for each shed. Treatment history was a binary variable recorded at the flock level indicating the presence or absence of flock-level medication at any time during the period in lay, without specification of the type, frequency and reason for treatment.

Production data included the count of birds entering each shed on the date of placement and the number of eggs produced each day. Eggs were categorised as either oversized, floor, dirty or suitable for sale. Shed managers provided a count of the number of bird deaths that occurred on each day of the follow-up period allowing flock size for each day to be calculated by subtracting the running total of bird deaths from the initial bird count on placement. Shed-level data included the minimum and maximum daily temperature (in degrees Celsius), the average volume of water consumed per bird (in millilitres) and the average weight of feed offered per bird per day (in grams). At each farm site weather stations (Digitech, Model: XC0370) were installed at a central location allowing minimum and maximum daily temperature, humidity, and rainfall to be recorded for the duration of the study.

Semi-structured interviews were conducted with flock managers at the end of each flock's follow-up period to record details of routine management practices, the number of staff that worked with the flock and qualitative assessments of hen behaviour. Details of the questions asked during the flock manager interviews are provided in Appendix A.

2.2. Behaviour assessments

A flight distance test and novel object test were used to assess behavioural responses of birds to humans in 49 of the 84 study flocks between 45 and 55 weeks of age. Not all flocks were tested due to COVID-19 restrictions that were in place at the time of data collection. For the flight distance test observations were recorded using action cameras (GoPro Hero 7 Black; California, USA) mounted on a chest strap to the investigator. The flight distance test, adapted from Cransberg et al. (2000) and Raubek et al. (2007), required the investigator (wearing either disposable overalls or a farm uniform in accordance with the specific farm's biosecurity protocols) to move through the shed and outdoor range in a standardised manner, proceeding at a rate of one step per second and stopping every 20 steps for 30 seconds. Based on digital images recorded by the action camera, the number of hens within a semi-circular area extending 1.25 m in front of the investigator was measured at every step while the investigator was moving and every 5 seconds while the investigator was stationary. Results for the flight distance test were expressed as the average number of hens within 1.25 m of the investigator for movement and stationary phases for inside the shed and the outdoor range. The novel object test, adapted from Donaldson and O'Connell (2012), involved placing a child's toy (a rainbow-coloured wooden stacker of height 22 cm and width 12 cm) at three different locations inside the shed and in the outdoor range. Once the toy had been placed, the investigator moved approximately 5 m away. From digital video recordings, the time (in seconds) taken for hens to come within 10–40 centimetres and to peck at the object and the average number of hens within 40 centimetres of the object was measured at 30-second intervals for a period of five minutes after each placement. Results for the novel object test were expressed as the average time taken for birds to come within 10–40 centimetres of the object and the average number of birds within 40 centimetres of the object at each of the ten 30 second intervals over the 5-minute observation period.

2.3. Smothering records

A bird was defined as having died from smothering if either it had no obvious signs of illness or injury (e.g., diarrhoea, discharge from the respiratory tract or discolouration of the skin) in an area where there was evidence of a recent disturbance of the litter consistent with piling, where a bird was found among a pile of two or more dead birds or if it was witnessed as a smothering death by flock management staff.

For each day of the study period flock managers, using sheets specifically designed for the study, recorded information about birds that died from smothering. Details included the date and time (if known) of the smothering event, the name of the staff member who made the observation, the location of the event, and the number of birds that had died. A shed map was developed to allow staff to record smothering locations inside and outside each shed (Fig. 1). This provided smothering location details down to an accuracy of approximately 2–5 m (for the longer of the two axes of a shed) and 8–30 m (for the shorter of the two axes).

Communication with farm staff took place at least once every two weeks throughout the follow-up period, initially through in-person meetings (prior to the implementation of COVID-19 travel restrictions in March 2020) and later by phone or teleconference. Data pertaining to production and records of smothering incidents were collected and transcribed into a relational database (Microsoft Access, Microsoft Corporation, Redmond, USA) by members of the research team.

Each of the variables recorded during the study period are listed and described in Table 1. Some variables (e.g., indoor and outdoor average temperature) were recorded daily while others (e.g., flock manager's general observations on bird behaviour and treatment history) were recorded on a single occasion at the end of the follow-up period. While we acknowledge that the variables relating to the entire production cycle were coarse (i.e., insensitive) indicators of smothering risk, our reasoning for including them (the presence or absence of health events, in particular) was to determine if flock managers' perceptions of flock characteristics showed any association with the magnitude of smothering incidence risk.

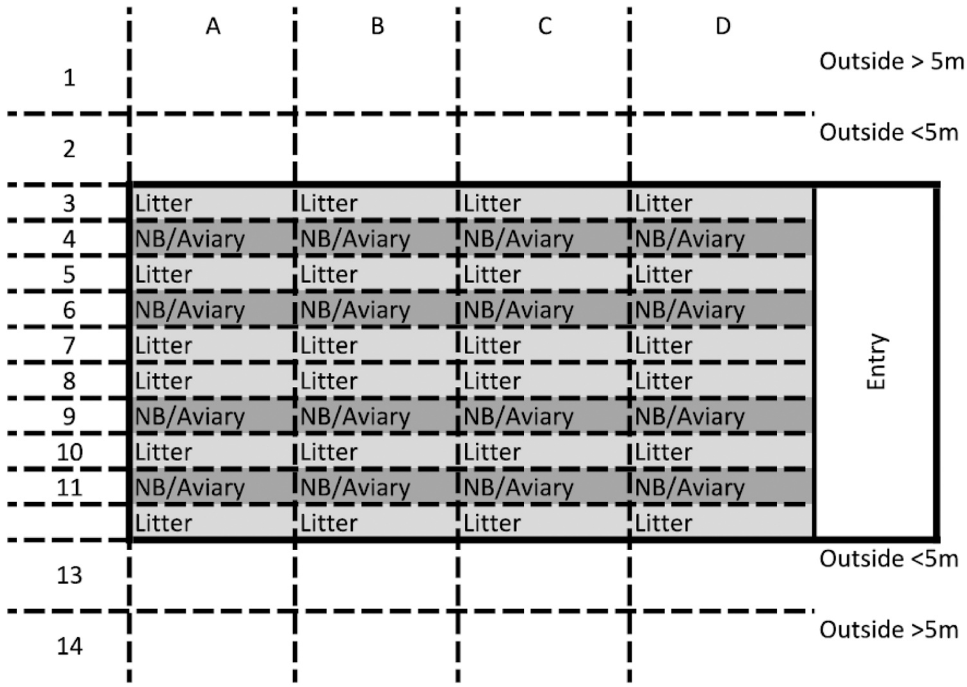

Fig. 1. Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Example of smothering logbook showing the grid system used to identify smothering locations. The long axis of the shed was divided into four parts (A, B, C, and D) and the short axis of the shed was divided into ten equal parts (3−12). The designated area on the outdoor range was marked as within (2 and 13) and outside (1 and 14) 5 m from the wall of the shed.

2.4. Statistical analyses

A directed acyclic graph (DAG) was developed to document hypothesised causal relationships between each of the candidate explanatory variables and smothering risk (Fig. 2). We used DAGitty (Textor et al., 2011) to construct the DAG and to determine sufficient adjustment sets (a set of variables included in a model to control for confounding) for each of the explanatory variables of interest. Two sets of Cox proportional hazards regression models were developed: the first for indoor smothering and the second for outdoor smothering events. Each model quantified the association between a given explanatory variable and the daily hazard of death from smothering, controlling (if necessary) for the sufficient adjustment set identified using DAGitty.

Because the results of the flight distance and novel object tests were each comprised of several variables, a principal component analysis (PCA) was carried out to return a single, composite summary score for each test. Prior to the PCA analyses, each of the outcome measures for each test were inverse $\log_{10}$ or square root transformed to achieve normality. Since the Pearson correlation coefficients between each of the outcome measures were greater than 0.3, the Kaiser-Meyer-Olkin values exceeded the recommended value of 0.6 and Bartlett's Test of Sphericity was statistically significant at the alpha level of 0.05, the data were assessed as being suitable for PCA. The PCA analyses returned, for each behaviour test, a summary numeric value centred at zero with values greater than zero representing a lower level of fear (in the case of the flight distance test) and greater fear of novelty (in the case of the novel object test).

The timing of flock-level smothering events in relation to days in lay and calendar date were quantified using survival analyses (Cox, 1972; Kaplan and Meier, 1958). The outcome of interest for our analyses was the calendar date on which a smothering event occurred. The number of birds that were still present in their respective flock at the end of the period in lay were treated as censored observations. Birds that died for reasons not related to smothering were censored on their recorded date of death. Due to the relatively large number of birds in each flock, numeric weights representing the number of birds experiencing the event of interest on a given day were used. Similarly, numeric weights were assigned to the number of birds present at the end of the period in lay that were still alive. For birds that were still alive at the end of the period in lay, the weight assigned was equal to the number of birds that entered the shed on the date of placement minus the total number of recorded smothering deaths.

The timing of smothering deaths in relation to the number of days in lay and calendar date was quantified using the Kaplan-Meier method (Kaplan and Meier, 1958). Kaplan-Meier estimates of survival are plotted with time on the horizontal axis and the corresponding proportion of surviving individuals at each time point on the vertical axis. Kaplan-Meier survival curves are presented as a series of horizontal steps, which are ever-declining in magnitude. The slope of the Kaplan-Meier survival curve indicates how quickly the event of interest is occurring over time. The vertical position of the Kaplan-Meier estimate at the end of the follow-up period quantifies the proportion of the group initially at risk that did not experience the event of interest.

Analyses were carried out comparing time to smothering death for birds with and without a given characteristic hypothesised to influence smothering risk (explanatory variables). The similarity of the Kaplan-Meier survival curves across levels of a given explanatory variable was tested using the log-rank statistic.

A key assumption of the Cox proportional hazards model is the proportionality of hazards. Accordingly, the effect of an explanatory variable on the outcome of interest does not change over time, that is, the hazards for each of the levels of an explanatory variable are proportional. This allows us to make statements about a given explanatory variable either increasing or decreasing the daily hazard of smothering by a stated amount (the hazard ratio). To verify the proportional hazards assumption, we plotted the scaled Schoenfeld residuals from the Cox model as a function of follow-up time. The Pearson product-moment correlation between the scaled Schoenfeld residuals and time was calculated and the hypothesis of no correlation between the two variables was assessed using a χ^2 test statistic. To account for the violation of the proportional hazards assumption at the organisation level, we stratified our Cox model by organisation (Kleinbaum and Klein, 2012).

Table 1
Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Source and description of each of the candidate explanatory variables included in this study.

Variables	Data source	Description
Flock:		
Breed	Farm records	Hy-line Brown, ISA Brown and mixed.
Flock number	Farm records	First or second flock within the same shed in the study.
Number of birds placed	Farm records	Number of birds in the flock at placement.
Weather:[a]		
Temperature	Weather station	Outside average daily temperature (°C).
Humidity	Weather station	Average daily humidity (%).
Barometric pressure	Weather station	Maximum daily atmospheric pressure (kPa).
Precipitation	Weather station	Daily precipitation (mm).
Staff:		
Staff per shed	Interview	The number of staff that worked with the flock daily.
Shed walks per day	Interview	Average number of full floor walks per day.
Egg production:		
Eggs per bird per day	Farm records	Total number of eggs per bird per day, recorded daily.
Floor eggs per bird per day	Farm records	Number of floor eggs per bird per day, recorded daily.
Inside shed environment:		
Temperature[b]	Farm records	Indoor average daily shed temperature (°C).
Water	Farm records	Average volume of water (mL) consumed (recorded daily).
Feed	Farm records	Average weight of feed offered (g) per bird recorded daily.
Water-to-feed ratio	Farm records	The feed offered to water consumed ratio, calculated daily using farm records.
Shed:		
Rearing shed type	Farm records	Type of rearing system (floor, aviary or Jump Start).
Production shed type	Farm records	Type of production system (flat deck or aviary).
Pop hole length	Farm records	Length of pop hole (cm) per bird.
Light intensity	Interview	Shed light intensity (lux).
Health:		
Disease events	Interview	Occurrence of disease in the flock during production (yes - no).
Behaviour:		
Piling behaviour	Interview	Score of prevalence of piling behaviour, perceived by shed managers.
Fear of humans	Behaviour tests	Score for behavioural responses to a human observer.
Fear of novel object (inside)	Behaviour tests	Score for behavioural responses to a novel object inside the shed.
Fear of novel object (outside)	Behaviour tests	Score for behavioural responses to a novel object in the outdoor range.

[a]As measured by weather stations located on each of the study farms.
[b]This is the range of temperature outside of which the organisations take protective management measures (e.g., activate foggers or raise curtains).

To account for the lack of independence in the data arising from individual birds clustered within flocks within sheds (Table 2), a Gaussian distributed frailty term was introduced to account for unobserved smothering risk at the shed level.

Our Kaplan-Meier survival curves and Cox proportional hazards models were developed using the contributed survival and coxme packages (Therneau and Grambsch, 2001) in R version 4.3.2 (R Core Team, 2024).

3. Results

Table 2 describes the hierarchical structure of the data gathered during this study. A total of 43 sheds were used for egg production with the number of sheds per organisation ranging from 9 to 18. Eighty-four flocks were followed with the number of flocks per shed ranging from 1 to 2. Throughout the follow-up period, a total of 287,706 deaths were recorded among the 2329,463 birds placed, a mortality incidence risk of 12 deaths for every 100 birds placed. Of the 287,705 deaths, 46,639 were due to smothering returning a smothering mortality incidence risk of 2 deaths for every 100 birds placed. The smothering mortality incidence rate was 183 deaths per 10,000 bird-years at risk (Chowdhury et al., 2024). This means that over a 12-month period in the order of $4 \times 183 = 732$ smothering deaths could be expected in a shed in which 40,000 birds were placed.

Of the 2471 smothering events recorded during the follow-up period, 161 (6.5 %) involved the death of a single hen, while the remaining 2310 events resulted in the death of two or more hens. Of the 161 single-hen smothering deaths 69 (10 %) occurred inside the shed, 25 (3 %) occurred outside the shed, and 67 (7 %) took place in locations that were not recorded.

Details of each of the continuously distributed candidate explanatory variables included in this study are provided in Table 3. The median number of birds per flock at the time of placement was 20,100 (Q1 19,300; Q3 22,300). The median number of days in lay (from the day of placement in the production shed to the last observation day) for each flock was 421 (Q1 407; Q3 447) days. Across the three organisations birds produced a median of 0.88 (Q1 0.82; Q3 0.92) eggs per bird per day. The PCA (Table 4) returned, for the novel object and flight distance test, a summary numeric value centred at zero with values greater than zero representing a greater fear of novelty (in the case of the novel object test) and a lower level of fear (in the case of the flight distance test). Three components, all with eigenvalues greater than one, were extracted from the PCA and are presented in Table 4.

Fig. 3 shows the cumulative proportion of birds to experience death by smothering as a function of days since flock placement, stratified by organisation. Fig. 3 shows a marked difference in smothering risk firstly as a function of days in lay and by organisation (log-rank test statistic 1745; *df* 2; $p < 0.01$). Fig. 4 shows the cumulative proportion of birds smothered as a function of days since flock placement, by breed and by organisation (log-rank test statistic 2708; *df* 2; $p < 0.01$).

Tables 5 and 6 present the crude and adjusted hazard ratios for indoor and outdoor smothering deaths, respectively. For birds housed in aviary sheds the daily hazard of indoor and outdoor smothering deaths was 4.0 (95 % CI 1.7–9.7) and 2.2 (95 % CI 1.2–4.1), respectively, compared with birds housed in flat deck sheds (Tables 5 and 6). Increases in the number of shed walks per day by shed staff decreased outdoor smothering risk. On days when there were less than or equal to two shed walks per day the daily hazard of indoor smothering deaths was increased by a factor of 1.2 (95 % CI 1.2–1.3) compared with days when greater than or equal to four shed walks were carried out (Table 6). There was no association between the number of shed walks per day and the daily hazard of indoor smothering (Table 5).

Increases in daily water-to-feed ratio decreased both indoor and outdoor daily smothering hazard. For days when water to water-to-feed ratio was greater than two the daily hazard of indoor smothering deaths was 0.81 (95 % CI 0.77–0.86) times that of days when the water-to-feed ratio was less than two (Table 4). A similar trend was observed for outside smothering deaths (0.85, 95 % CI 0.81–0.89, Table 6).

Birds that had an estimated low fear of humans (i.e., flocks with flight distance test scores of greater than or equal to zero) had a 1.8 (95 % CI 1.5–2.3) and 1.4 (95 % CI 1.2–1.6) times increase in the daily hazard of indoor and outdoor smothering deaths, respectively (Tables 5 and 6). Birds with an estimated low fear of novel objects indoors had a 2.9 (95 % CI 2.5–3.3) and 1.3 (95 % CI 1.3–1.5) times increase in the daily hazard of indoor and outdoor smothering deaths, respectively (Tables 5 and 6). Birds with low fear of novel objects outdoors had a similar pattern (2.9, 95 % CI 2.5–3.3 and 1.4, 95 % CI 1.3–1.5 respectively for indoor and outdoor smothering deaths, Tables 5 and 6).

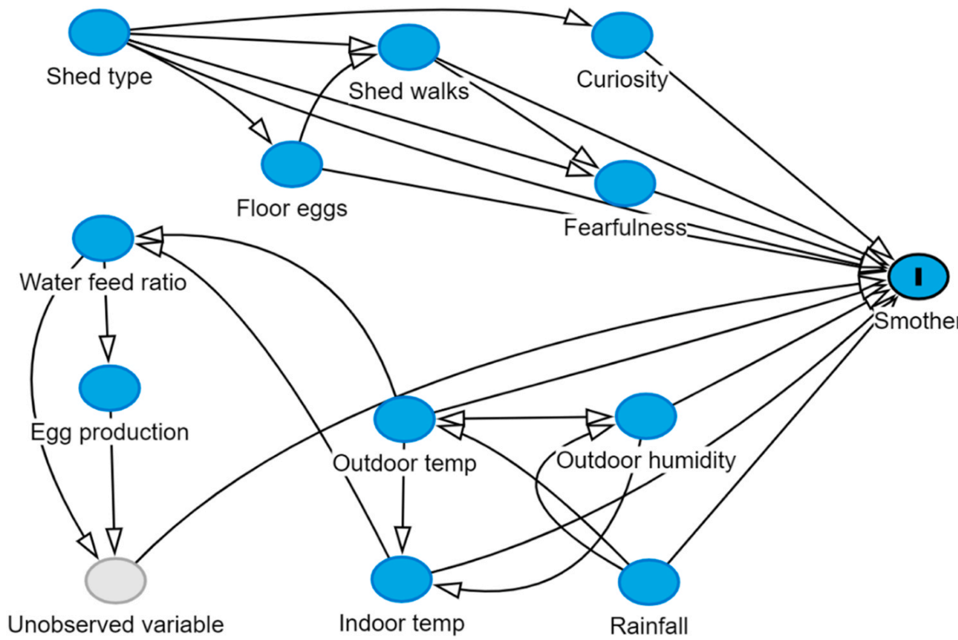

Fig. 2. Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Directed acyclic graph showing putative causal paths linking explanatory variables to the daily probability of hen death due to smothering.

Table 2
Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Hierarchical structure of the data for the three organisations included in this study.

Level	n	The number at next highest level		
		Median	Min	Max
Organisation	3			
Sheds	43	16[a]	9	18
Flocks	84	2	1	2
Birds	2329,463	20,000	16,000	45,000

[a] Interpretation: The median number of sheds per organisation (i.e., the next highest level) was 16.

The daily hazard of smothering mortality was increased on warm, humid and rainy days. Indoor smothering death hazard was increased by a factor of 1.8 (95 % CI 1.6–1.9) on days when outdoor daily average temperatures were between 10 and 27 ℃ compared with days when outdoor daily average temperatures were less than 10 ℃ (Table 5). For days when outdoor daily temperature was greater than 27 ℃, the daily hazard of indoor smothering deaths was increased by a factor of 2.5 (95 % CI 2.2–2.8, Table 5). The outdoor smothering deaths hazard ratio for days on which the outdoor average daily temperature was between 10 and 27 ℃ was similar to indoor smothering deaths, 2.4 (95 % CI 2.2–2.6, Table 6). Days when outdoor average daily temperatures were greater than 27 ℃ were not associated with an increase in the hazard of outdoor smothering (0.97, 95 % CI 0.84–1.1, Table 6).

Rainy days on which outdoor daily average humidity was greater than or equal to 70 % and were associated with a 3.7 (95 % CI 3.5–3.9) fold increase in the daily hazard of indoor smothering deaths, compared with days when outdoor daily average humidity was less than 70 % and no rain (Table 5). A similar trend was apparent for outdoor smothering deaths (2.0, 95 % CI 1.9–2.0, Table 6).

4. Discussion

We identify marked differences in smothering risk by organisation with Organisation 2 losing 2.4 birds to smothering for every 100 birds placed compared with 1.5 smothering deaths per 100 for Organisation 1 (Fig. 3). Smothering risk varied by breed with both ISA Brown and mixed breed flocks having a lower risk of smothering compared with Hy-Line Brown (HB) flocks (Fig. 4). Organisation 1 had the lowest smothering risk overall and, while all organisations had HB birds, Organisation 1 was the only organisation that had both ISA and mixed breed flocks. Thus, it is more than likely that the breed differences identified here were confounded by organisation. Even so, breed differences have

Table 3
Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Descriptive statistics for each of the continuously distributed variables collected as part of this study.

Variable	n	Mean (SD)	Median (Q1, Q3)	Min, Max	Missing[a]
Number of birds placed (× 1000)	34,974	22 (4.6)	20 (19, 22)	16, 41	-
Number of days in lay[b]	84	416 (54)	421 (407, 447)	162, 503	-
Egg per bird per day[c]	34,974	0.81 (0.22)	0.88 (0.82, 0.92)	0, 9	790
Water feed ratio	34,974	1.6 (0.5)	1.6 (1.4, 1.8)	0.01, 23	1333
Average indoor temperature (°C)	34,974	20 (4)	21 (18, 24)	7, 35	1316
Average outdoor temperature (°C)	34,974	18 (6)	18 (13, 22)	4, 34	3758
Average outdoor humidity (%)	34,974	67 (15)	69 (58, 78)	13, 99	3758
Daily rainfall (mm)	34,974	2 (7)	0 (0, 0)	0, 171	3771

[a]Number of production days where no details were entered for the respective variable.
[b]Total number of days calculated at the flock level.
[c]Total number of eggs per bird per day calculated at the flock level.

Table 4
Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Pattern matrix from the principal components analysis with oblique rotation, illustrating the three-factor solution for test items from the novel object test and the flight distance test.

Test, variable	NOT_{Indoor}
Novel object test:	
Indoor	
Latency to peck	0.885
Latency to 10 cm	0.921
Latency to 40 cm	0.727
Mean number of hens within 40 cm (2 min)	0.880
Mean number of hens within 40 cm (5 min)	0.839
Outdoor	
Latency to peck	0.827
Latency to 10 cm	0.735
Latency to 40 cm	0.638
Mean number of hens within 40 cm (2 min)	0.969
Mean number of hens within 40 cm (5 min)	0.969
Flight distance test:	
Indoor	
Mean number of hens within 1.5 m, movement phase	1.023
Mean number of hens within 1.5 m, stationary phase	0.838
Outdoor	
Mean number of hens within 1.5 m, movement phase	−0.733
Mean number of hens within 1.5 m, stationary phase	−0.554

been observed in the piling behaviour of laying hens (Winter et al., 2021) and in the occurrence of nest box smothers reported by flock managers (Rayner et al., 2016). Also, de Haas et al. (2013) found differences in the occurrence of smothering between Dekalb white and ISA flocks. Acknowledging that we were unable to definitively distinguish organisation and breed differences in smothering risk in this study, further investigations into breed effects on smothering risk are warranted.

The daily hazards of indoor and outdoor smothering death for birds housed in aviary sheds were respectively four times and two times higher compared to birds housed in flat-deck sheds. Similar to the breed issue discussed above, we were not able to distinguish between organisation-level effects and the effect of three differently designed flat-deck sheds (i.e., fully slatted floors, partially slatted floors with indoor scratch areas and fully slatted floors with a winter garden). A difficulty when conducting observational epidemiological research in intensively managed poultry is that large numbers of birds within an organisational grouping (e.g., farm or flock) share identical exposure variables, such as breed, shed type and shed ventilation method, making it difficult to separate specific exposure effects (e.g., breed, shed type, shed ventilation method) from the effect of the organisational unit (farm, flock). When designing observational studies in intensively-managed commercial poultry populations, a useful rule of thumb is to aim to collect data from as many primary sampling units (farms, flocks) as possible, as opposed to collecting data from large numbers of secondary sampling units (birds) within a smaller number of primary sampling units. The inclusion of shed type in our analyses was deemed

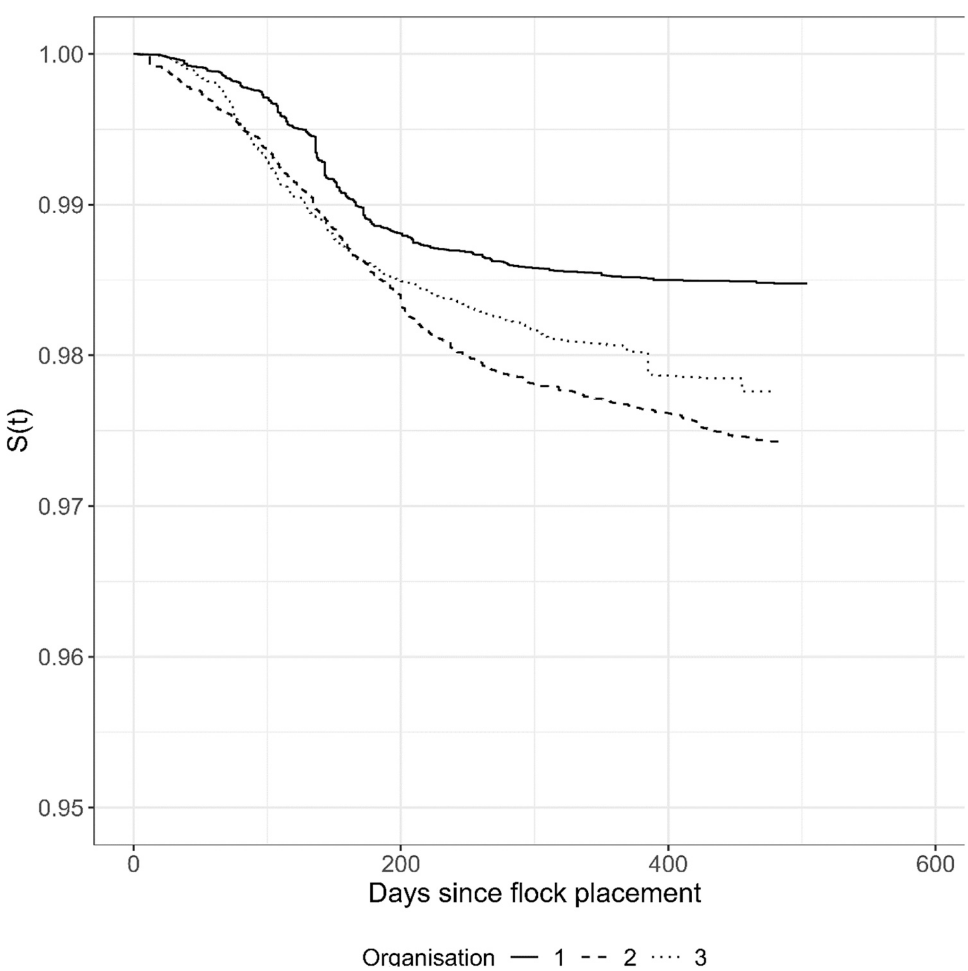

Fig. 3. Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Kaplan-Meier survival curves showing the cumulative proportion of birds to experience death by smothering as a function of days since flock placement, by organisation. Differences in time to event by organisation were statistically significant at the alpha level of 0.05 (log rank test statistic 1745; *df* 2; $p < 0.01$).

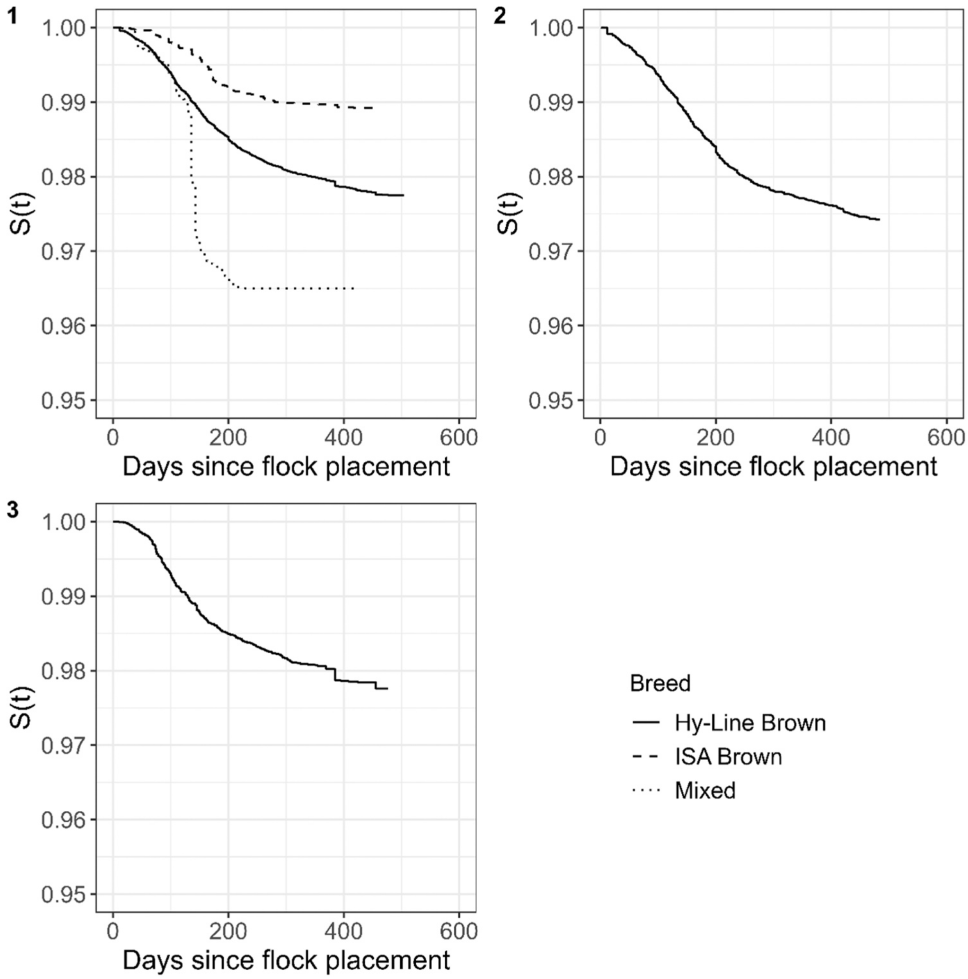

Fig. 4. Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Kaplan-Meier survival curves showing the cumulative proportion of birds to experience death by smothering as a function of days since flock placement and breed (Hy-Line Brown, ISA Brown and mixed): (1) Organisation 1; (2) Organisation 2; (3) Organisation 3.

necessary based on insights from previous research, such as studies conducted by Sherwin et al. (2010) and Brantsaeter et al. (2018). These studies highlighted the significant impact that shed type can have on the welfare of laying hens, underscoring the relevance of shed design in the broader context of poultry welfare.

On days when the outdoor average daily temperature was greater than 27 ℃, there was a 2.5 (95 % CI 2.2–2.8) fold increase in the daily hazard of indoor smothering deaths compared with days when outdoor daily average temperatures were less than 10 ℃ and a 0.97 (95 % CI 0.84–1.1) fold decrease in the daily hazard of outdoor smothering deaths compared with days when outdoor average daily temperatures were less than 10 ℃. These findings are biologically plausible since when conditions are hot, birds are likely to prefer to stay inside the shed leading to a higher risk of smothering. It would be of interest to determine details of how flock management was changed on hot days since these changes (e.g., decreasing the number of shed walks and closing pop holes) may have unintentionally increased smothering mortality risk. Bright and Johnson (2011b) proposed a correlation between smothering (panic and creeping smothers) and fluctuations in temperature. During the course of the study flock managers cited instances of hens clustering together during dust-bathing activities when conditions were sunny, as well as incidents of birds congregating and subsequently suffocating in sunlit areas indoors on the litter or outdoors on the range. Indoor smothering events were less likely to occur on days when the indoor average temperature was high ($\geq$27 ℃) compared to days with lower indoor average temperatures (HR 0.91, 95 % CI 0.81–1.0). These findings support findings of Pereira et al. (2020) who reported that when temperatures were > 27 ℃ birds tended to stay more spread out and were less likely to aggregate into groups. If modifying shed temperatures was contemplated as an intervention to reduce smothering risk, producers are advised to consider the impact of that change on the entire production system, with higher temperatures increasing the risk of conditions such as spotty liver disease as well as having negative effects on egg production and egg quality.

The increased risk of smothering associated with low water-to-feed ratios found in this study is contrary to the findings of Herbert et al. (2021) who, in a study of a single flock of 12,000 birds over 35 days, identified a positive association between piling extent (the maximum number of birds involved in a piling event) and water-to-feed ratio. Differences in outcome variables, that is the number of birds involved in piling events (Herbert et al., 2021) and smothering deaths in this study make it difficult to compare the findings reported here with those of Herbert et al. (2021).

An observation by flock managers and members of the research team during the follow-up period was that birds were attracted to shards of light within a shed. This was also mentioned by Winter et al. (2022) in relation to UK producer comments. This, together with synchronicity of behaviour, is hypothesised to lead to the formation of piles and death from smothering. The context in which piles lead to smothering is unknown but, for example, physically weak or stressed birds (such as those that are high producing or diseased) may be more susceptible to suffocation in piles. Both inside and outside smothering risk was increased in

Table 5
Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Regression coefficients and their standard errors from a stratified, mixed-effects Cox proportional hazards model of factors associated with indoor smothering risk.

Explanatory variable	Smothered	Hazard ratio (95 % CI)		Adjustment set
	(*n*)[a]	Crude	Adjusted[b]	
Shed type:				
Flat-deck	3090 (327)	Reference	Reference	-
Aviary	9661 (601)	4.0 (1.7–9.7)	-	-
Random effect variance[c]	1.4771			
Shed walks per day[d]:				
≥ 4	3309 (256)	Reference	Reference	
3	4953 (363)	1.1 (1.0–1.1)	0.89 (0.83–0.95)	Shed type, floor eggs
≤ 2	4272 (276)	1.2 (1.1–1.3)	0.91 (0.84–0.99)	Shed type, floor eggs
Random effect variance	1.8858			
Water to feed ratio:				
< 2	10847 (768)	Reference	Reference	
≥ 2	1824 (119)	0.81 (0.77–0.86)	0.80 (0.75–0.85)	Indoor temp, outdoor temp
Random effect variance	2.3487			
Fear of humans:				
High	2692 (257)	Reference	Reference	
Low	4529 (317)	0.99 (0.88–1.1)	1.8 (1.5–2.3)[e]	Shed type, shed walks
Random effect variance	5.4549			
Fear of novel objects (indoor):				
Low	3925 (300)	Reference	Reference	
High	2852 (236)	2.4 (2.1–2.7)	2.4 (2.1–2.7)	Shed type
Random effect variance	2.3752			
Fear of novel objects (outdoor):				
Low	1961 (164)	Reference	Reference	
High	4816 (371)	2.9 (2.5–3.3)	2.9 (2.5–3.3)	Shed type
Random effect variance	2.6496			
Indoor average temp (°C):				
< 27	12343 (870)	Reference	Reference	
≥ 27	349 (22)	0.79 (0.71–0.89)	0.91 (0.81–1.0)	Outdoor temp, humidity, rainfall
Random effect variance	2.5560			
Outdoor average temp (°C):				
< 10	955 (97)	Reference	Reference	
10—27	9576 (700)	1.1 (1.0–1.1)	1.8 (1.6–1.9)	Outdoor humidity, rainfall
> 27	485 (34)	0.83 (0.74–0.93)	2.5 (2.2–2.8)	Outdoor humidity, rainfall
Random effect variance	2.5506			
Outdoor average humidity (%) and rainfall (mm):				
< 70 and 0	3230 (393)	Reference	Reference	
≥ 70 and 0	3993 (258)	2.4 (2.3–2.5)	-	-
< 70 and > 0	449 (42)	1.8 (1.6–2.0)	-	-
≥ 70 and > 0	3206 (138)	3.7 (3.5–3.9)	-	-
Random effect variance	2.5937			

[a] × 1000,000 bird-days.
[b] The effect estimates, hazard ratio, and confidence intervals provided here reflect the overall impact, derived from the minimum adjustment sets guided by the directed acyclic graph shown in Fig. 2.
[c] Variance of the shed level random effect term.
[d] The number of times the shed was 'walked' (i.e., checked) by staff each day.
[e] Interpretation: After adjusting for the effect of shed type and the number of shed walks per day, birds that had a lower fear of humans (i.e., flocks with high human test scores) had 1.8 (95 % CI 1.5–2.3) times the daily hazard of smothering compared with birds that had a higher fear of humans.

flocks with a lower level of fear of humans, that is when more birds remained near the investigator in the movement and stationary phases of the flight distance test. For both indoor and outdoor smothering deaths adjustment for the effect of shed type and the number of shed walks per day reversed the direction of the association between the flight distance test score and smothering hazard, an example of Simpson's paradox (Simpson, 1951), showing that the number of shed walks per day was a relatively strong confounder in the association between human response score and smothering risk. Conversely, both indoor and outdoor smothering risk was increased in flocks with a higher level of fear of novel objects in both indoor and outdoor settings (Table 5 and Table 6). Fear of or attraction to stimuli introduced into the shed and the range may result in rapid bird dispersal or approach, respectively which might lead to piling and consequent smothering deaths.

In studying behavioural responses of farm animals, including poultry, to humans numerous studies have adopted a functional view and used the avoidance behaviour of the animal to an approaching experimenter or, conversely, the approach behaviour of the animal to a stationary experimenter to assess fear of humans: see, for example, Hemsworth and Coleman (2011), Forkman et al. (2007) and Waiblinger et al. (2006). While it is recognised that there are likely to be competing motivations such as fear and exploration in such behavioural tests and that it is generally agreed that very high levels of fear inhibit all other motivational systems including exploration (Feenders et al., 2011) the rationale in these tests is that reduced approach behaviour and conversely increased avoidance behaviour are likely to be indicative of increased fear (Hemsworth and Coleman, 2011). Edwards et al. (2007) assessed the fear of novelty in farm animals, including poultry, by observing the behavioural responses of animals to a novel object, while Forkman et al. (2007) provided evidence supporting the validity of such tests as indicators of fearfulness in animals.

The shed-level random effect term included in the Cox proportional hazards model is a metric that quantifies unmeasured, shed-level characteristics associated with smothering risk. Used in this way a random

Table 6
Risk factors for smothering in three commercial free-range layer poultry farms, Australia 2019–2022. Regression coefficients and their standard errors from a stratified, mixed-effects Cox proportional hazards model of factors associated with outdoor smothering risk.

Explanatory variable	Smothered	Hazard ratio (95 % CI)		Adjustment set
	(*n*)[a]	Crude	Adjusted[b]	
Shed type:				
Flat-deck	4778 (327)	Reference	Reference	-
Aviary	14504 (601)	2.2 (1.2–4.1)	-	-
Random effect variance[c]	0.7572			
Shed walks per day[d]:				
≥ 4	4756 (256)	Reference	Reference	-
3	6953 (363)	0.92 (0.88–0.97)	0.98 (0.93–1.0)	Shed type, floor eggs
≤ 2	7438 (276)	0.98 (0.94–1.0)	1.24 (1.2–1.3)	Shed type, floor eggs
Random effect variance	1.0696			
Water to feed ratio:				
< 2	15467 (768)	Reference	Reference	-
≥ 2	3342 (119)	0.92 (0.89–0.96)	0.85 (0.81–0.89)	Indoor temp, outdoor temp
Random effect variance	2.0311			
Fear of humans:				
High	4658 (257)	Reference	Reference	-
Low	8448 (317)	0.90 (0.80, 1.0)	1.4 (1.2–1.6)[e]	Shed type, shed walks
Random effect variance	2.1088			
Fear of novel objects (indoor):				
Low	7634 (300)	Reference	Reference	-
High	4815 (236)	1.3 (1.3–1.5)	1.3 (1.3–1.5)	Shed type
Random effect variance	1.8927			
Fear of novel objects (outdoor):				
Low	4666 (164)	Reference	Reference	-
High	7783 (371)	1.4 (1.3–1.5)	1.4 (1.3–1.5)	Shed type
Random effect variance	2.4332			
Indoor average temp (°C):				
< 27	18731 (870)	Reference	Reference	
≥ 27	248 (22)	0.53 (0.47–0.60)	0.57 (0.48–0.64)	Outdoor temp, humidity, rainfall
Random effect variance	2.0950			
Outdoor average temp (°C):				
< 10	968 (97)	Reference	Reference	
10-–27	16341 (700)	2.1 (1.9–2.2)	2.4 (2.3–2.6)	Outdoor humidity, rainfall
> 27	278 (34)	0.73 (0.64–0.84)	0.97 (0.84–1.1)	Outdoor humidity, rainfall
Random effect variance	2.0146			
Outdoor average humidity (%) and rainfall (mm):				
< 70 and 0	7226 (393)	Reference	Reference	
≥ 70 and 0	4366 (258)	1.3 (1.2–1.3)	-	-
< 70 and > 0	1543 (42)	1.7 (1.6–1.8)	-	-
≥ 70 and > 0	4452 (138)	2.0 (1.9–2.0)	-	-
Random effect variance	2.0514			

[a] × 1000,000 bird-days
[b] The effect estimates, hazard ratio, and confidence intervals provided here reflect the overall impact, derived from the minimum adjustment sets as guided by the directed acyclic graph shown in Fig. 2.
[c] Variance of the shed level random effect term.
[d] The number of times the shed is 'walked' (i.e., checked) by staff each day.
[e] Interpretation: After adjusting for the effect of shed type and number of shed walks per day, birds that had a lower fear of humans (i.e., flocks with high human test scores) had 1.4 (95 % CI 1.2–1.6) times the daily hazard of smothering compared with birds that had a greater fear of humans.

effect term (by definition) is centred at zero and the numeric estimate of a shed's random effect term itself has no inherent meaning: inference from the random effect term comes from its variability. In situations where a shed-level random effect term has a large variance, our inference is that there is wide variation in shed-level smothering risk that has not been quantified by the explanatory variables included in the model. In situations where a shed-level random effect term has a relatively small variance (i.e., individual shed random effect terms are tightly clustered around zero when plotted as a frequency histogram), the inference is that the contribution of unmeasured shed-level effects on smothering risk is relatively small—that is, the (fixed) explanatory variable effects included in the model account for most of the variation in smothering risk. The variance of the shed-level random effect terms from the Cox proportional hazards models developed for indoor smothering risk ranged from 1.48 to 5.45 and for outdoor smothering events the range was from 0.76 to 2.43 (Tables 5 and 6). These metrics were relatively large indicating that unmeasured shed-level effects were determinants of smothering risk, in addition to the adjustment variables included in each of the models listed in Tables 5 and 6. A useful area of future investigation would be to compare the management practices and physical characteristics of 'high-risk' sheds (i.e., those with a shed-level random effect term greater than zero) with those of 'low-risk' sheds (those with a shed-level random effect term less than zero) to identify some of the more subtle determinants of smothering risk in free-range layer hens.

Care was taken in this study to document likely causal pathways for smothering and to identify sufficient adjustment sets based on the variable relationships documented in our directed acyclic graph (Fig. 2). This approach provided hazard ratio estimates that were believed to be unbiased estimates of the association between each risk factor and smothering risk (Töhilnnies et al. 2022). With this task completed, attention turned to assessment of causal effects. Shed type, fear of humans, fear of novel objects and outdoor average humidity and rainfall satisfied at least two of Hill's eight criteria (association strength, consistency, specificity, temporality, dose-response effect, plausibility, analogy and experimental evidence, Hill, 1965) supporting the inference that these risk factors were, in fact, component (but not necessary) causes of smothering (Rothman and Greenland, 2005).

5. Conclusion

This study identifies the variation in smothering risk by organisation, shed design, bird management, bird behaviour and weather conditions. Aviary sheds, lower water-to-feed ratios, a lower fear of humans, a higher level of fear of novel objects, and days with high outdoor humidity and rain, increased the smothering risk. Conversely, flat-deck sheds and higher water-to-feed ratios were associated with lower smothering risk.

This study provides useful insight into the possible determinants of smothering in Australian free-range layer hens, in particular risk factors that do not change over time (e.g., shed type) and those that change daily (e.g., weather conditions). Further investigations are required to determine the particular features of aviary sheds that increase smothering risk. Differentiation of the learned versus genetic contributions to the bird behaviour risk factors identified in this study warrants further investigation. Changing the way flocks are managed on humid and wet days is expected to be an intervention that can be implemented by shed managers relatively quickly and easily.

Declaration of Competing Interest

None of the authors has any financial or personal relationships that could inappropriately influence or bias the content of this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.prevetmed.2025.106568.

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